

Topic 6. ROBOTICS AND ARTIFICIAL INTELLIGENCE.

"I, on the other hand, am a finished product. I absorb electrical energy directly and utilize it with an almost one hundred percent efficiency. I am composed of strong metal, am continuously conscious, and can stand extremes of environment easily. These are facts which, with the self-evident proposition that no being can create another being superior to itself, smashes your silly hypothesis to nothing."
Isaac Asimov, "I, Robot."

<https://www.youtube.com/watch?v=fn3KWM1kuAw> (Do you love me? 3 min. 2021. Boston Dynamics)

DISCUSSION/SPEAKING PRACTICE

Artificial Intelligence

Please study the following links.

Audios:

<https://www.bbc.co.uk/programmes/w3ct1jg4> (BBC Business Daily, Killer Robots, August 17, 2021)

Texts:

- <https://futureoflife.org/background/benefits-risks-of-artificial-intelligence/>
- <https://suryamaddula.medium.com/types-of-ai-4ac1ee30e51c> (Types of AI, article, 2021)
- <https://www.nytimes.com/2022/09/02/technology/ai-artificial-intelligence-artists.html>
- https://obamawhitehouse.archives.gov/sites/default/files/whitehouse_files/microsites/ostp/NSTC/preparing_for_the_future_of_ai.pdf
- <https://www.washingtonpost.com/technology/2023/06/02/ai-taking-jobs/> (ChatGPT)
- <https://www.theguardian.com/commentisfree/2026/mar/03/anthropic-openai-pentagon-ethics>

Videos:

- <https://www.youtube.com/watch?v=gOsVjdZZCOg> Recycling plastics 4.0: Artificial intelligence and digital twin save resources (2021, 14 min)
- https://www.ted.com/talks/jeff_dean_ai_isn_t_as_smart_as_you_think_but_it_could_be (2021, 18 min)

Take time to explore the following key terms — understanding and using them will be essential for the upcoming discussion.

AI Terms for Discussion

Term	Definition
Anthropomorphism	The attribution of human traits to AI systems, often leading to misconceptions about their intelligence or intentions.
Artificial General Intelligence (AGI)/Strong AI	A theoretical form of AI with human-level cognitive abilities across any task or domain.
Artificial Narrow Intelligence (ANI)/Narrow AI/Weak AI	AI specialized in a single task or domain (e.g., spam filtering, image classification).
Artificial Superintelligence (ASI)	A hypothetical future AI that surpasses human intelligence in all aspects.
Bias in AI	Systematic errors or inaccuracies in AI algorithms due to flawed/biased data, assumptions, design choices or societal influences, leading to unfair or discriminatory outputs.
Cognitive Computing	AI systems designed to mimic human reasoning and understanding, often involving natural language and symbolic logic.
Computer Vision	A field of AI focused on enabling machines to interpret and understand visual data.
Conversational AI	AI designed for human-like conversation, including chatbots, voice assistants, and dialogue systems.
Data Mining	The process of discovering patterns and insights from large datasets, often using statistical or machine learning (ML) methods.
Data Scraping	Automated extraction of structured data from web pages, often used in AI to build datasets.
Deep Learning	A subset of ML using neural networks with many layers to learn complex representations.
Edge AI	AI computation performed on devices locally (e.g., smartphones, sensors), reducing latency and privacy risks.
Ethical AI	The study and application of ethical principles in AI development and deployment (e.g., fairness, transparency).
Explainable AI (XAI)	AI systems that provide understandable justifications for their decisions or actions.
Generative AI	AI that produces original content such as text, images, music, or code.
Hallucination (in AI)	When an AI model produces plausible-sounding but false or nonsensical output, especially in LLMs.
Human-in-the-Loop (HITL)	An AI design paradigm where humans are involved in training, validating, or overriding AI decisions.
Interactive AI	AI that engages in ongoing communication and learning through real-time interaction with users or environments.
Large Language Model (LLM)	A deep neural network trained on massive text data to generate and understand natural language.
Machine Learning (ML)	A core subfield of AI where models improve their performance by learning from data.
Multi-Modal AI	AI capable of processing and reasoning across different types of data (e.g., text, image, audio).
Natural Language Processing (NLP)	The field of AI focused on understanding, interpreting, and generating human language.
Personalized AI	AI systems that adapt to individual users' behaviors, preferences, and needs.

Predictive AI	AI systems that forecast future events or behaviors based on historical data.
Prompt Engineering	The practice of crafting inputs (prompts) to control the outputs of language models like GPT.
Reactive AI	The most basic form of AI that reacts to inputs without memory or learning capability.
Reinforcement Learning (RL)	An AI paradigm where agents learn optimal behavior through trial-and-error and rewards.
Responsible AI	A framework for developing AI systems with attention to fairness, accountability, and transparency.
Self-Aware AI/Sentient AI	A speculative form of AI that possesses consciousness or a sense of self (not yet achieved).
Self-Supervised Learning	A learning paradigm where the model generates its own labels from raw data, reducing the need for human annotation.
Sentiment Analysis	A type of NLP that detects emotional tone or attitude in text.
Symbolic AI	AI that uses symbols and logic (rather than statistical patterns) to represent and manipulate knowledge.
Transfer Learning	A technique where a model trained on one task is adapted for another, related task.
Zero-Shot Learning	The ability of an AI system to solve tasks without direct prior training, using general knowledge.

WATCH THE FOLLOWING VIDEO

<https://www.youtube.com/watch?v=tdUwWOZPn1M>

ICRA 2023 (2023, 15 min)

Describe the main functions/features/disadvantages of the following robots:

Useful vocabulary: a ‘**degree of freedom**’ in robotics usually refers to the number of movable joints of a robot.

1. Neo by 1X (Halodi Robotics)

2. Ameca by Engineered Arts

3. Jueying Lite 3 by Deep Robotics

4. X20 by Deep Robotics

5. Geminoid by Hiroshi Ishiguro

6. B1 by Unitree Robotics

7. Robot hands by Shadow Robot Company

8. Tita by Direct Drive

9. Aerial-AM by Imperial College London and EMPA

10. Robotic arm by Dyson Robotics Laboratory

11. Drones/quadcopters by Bitcraze

12. Richie by Pollen Robotics

TOPIC RELATED VOCABULARY

Choose **five-ten new words** and write down the definition in English next to the chosen words.

actuator	
articulation	
ball joint	
bipedal	
cantilever	
cartesian robot	
clamp	
dead man's switch	
dexterity	
envelope	
exoskeleton	
gantry	
gripper	
haptic interface	
master-slave	
manipulator	
payload	

pick-and-place cycle	
pronation	
quadrupedal	
servo	
slew rate	
supination	
spatial resolution	
swivel arm robot	
ultrasonic sensor	
velocity	
yaw	

Optional activity: Visit this website for more vocabulary

<https://robotsguide.com/learn/robotics-glossary>

READING I

BOTS IN THE DARK

Read the text and complete the tasks. There is a glossary at the end of each part of the text.

Today, robotic platforms are used by many first responders to scout disaster areas from the air or on the ground. But in mines, caves and other underground settings their use is very limited. In DARPA'S Subterranean Challenge teams from all over the world test new approaches to rapidly map, navigate, and search underground environments.

In a dark and damp abandoned coal mine outside of Pittsburgh, competitors from around the world gathered last August to see how robots, both ground and aerial, can best

locate humans and objects underground. The tunnel circuit, the first of three environments planned over the next two years, was part of the DARPA Subterranean Challenge.

Over the course of four days, 11 teams participated in simulated search and rescue missions, with early lessons learned that could have big implications for the military, first responders, and even space exploration.

DARPA placed 20 artefacts inside the coal mine, and teams were tasked to find as many of these as they could. The challenge was designed to "focus the technology on mobility, autonomy, networking, and perception," said Dr. Timothy Chung, a program manager at DARPA. "It's not about having technology that only works in one location at one time, but about having technology that is versatile and agile."

In February 2020, the robots will be tested in urban areas, and six months later it will be caves. "This doesn't give teams enough time to go back to the drawing board and completely rework their robots," notes Chung. "They have to be thinking about these different terrains from the first competition."

Each team was given 60 minutes in each of their four runs in the mines. To earn a point, teams had to find an artefact, identify it, locate it, and then report it back to the DARPA command post within an accuracy of five meters. Teams had had very little interaction with each other before convening in Pittsburgh, and ended up adopting a dizzying array of approaches to tackling this mission.

DRIVE, ROLL, WALK, FLY, FLOAT

There was the most variety between teams in the way they approached mobility in the Subterranean Challenge. Team Explorer, made up of members from Carnegie Mellon University and Oregon State University, won the circuit by deploying ground robots with large wheels. While the team had aerial robots as well, it leaned heavily on the wheeled robots because of their robustness.

"In the urban and cave environments in the next couple of circuits, there will be areas where the aerial vehicles will have an advantage," said Dr. Geoffrey Hollinger, assistant professor of mechanical engineering at Oregon State University, and one of the leads on Team Explorer. "But in the tunnel environment, many of the artefacts were accessible to the ground vehicles. I'm personally pretty happy so far, because I think we've covered the majority of the course with these robots. You'll see a wider array of robots from us in future circuits for sure."

Team PLUTO, made up of members from the University of Pennsylvania, Exyn Technologies, and Ghost Robotics, deployed both legged and flying robots. Its main focus was on making the systems agile so the robots could tackle the different terrains in all the circuits.

"The legged robot has a very high stance and can also swing its legs pretty high," said Dr. C.J. Taylor, professor of computer and information science at the University of Pennsylvania, and lead for Team PLUTO. "This lets it step over a lot of different things. Similarly, the flying robot has the ability to ascend vertically to find artefacts."

GLOSSARY

abandon, to	verlassen, aufgeben	first responder	Ersthelfer(in)
accessible	zugänglich, erreichbar	float, to	schweben

agile	beweglich, wendig	implication	Auswirkung
array	Reihe, Palette, Sortiment	legged	mit Beinen
artefact	Gegenstand	perception	Wahrnehmung
achieve, to	erreichen, schaffen	scout, to	erkunden, aufklären
cave	Höhle	stance	Haltung, Position
circuit	hier: Rundkurs, Rennstrecke	subterranean	unterirdisch
convene, to	zusammenkommen, sich versammeln	tackle, to	angehen, bewältigen
deploy, to	einsetzen	versatile	vielseitig, flexibel

Other teams, like CRETISE, made up of members from FLIR Systems and Neya Systems, deployed robots with treaded "arms" that boasted a 360-degree range that enable them to tackle very rough terrain. A version of this robot has already been sold to the military, according to David Weatherwax of FLIR, and a unique feature is that it can shift its weight forward when it climbs to prevent itself from falling backward.

Team NCTU, a self-funded team from the National Chiao Tung University in Taiwan, deployed blimps. "They have a much longer battery life than drones" said Yiwei Huang of NCTU, although the team admitted that manoeuvrability in the mines was a challenge. He said that for the next circuit, the team is considering adding some motors on the side so the blimp can change direction more easily and will probably make the blimps smaller so they fit better underground. Several teams mentioned that their drones had crashed while completing the circuit.

Aerial vehicles experienced both battery and payload constraints. They "get about 15 minutes of flight time," according to Dr. Sean Humbert, professor of robotics and system design at the University of Colorado Boulder, and the lead for Team MARBLE. "That basically gets you down the first hallway." The wheeled robots his team deployed "don't need a battery charge, and can run for the full hour." He said the "urban underground challenge next February is going to be very aerial vehicle focused. There are going to be stairs and large drop-offs. I'm thinking this will be like a metro station or something. I think a lot of the platforms that are having success in the mines won't necessarily be able to translate their success the same way in the next circuits".

CONNECTIONS AND AUTONOMY

Many teams focused on multi-robot coordination. This will only become more difficult in future competitions, said Hollinger of Team Explorer, "because we'll be thinking about larger environments, more 3D structures, and potentially more robots in these environments."

Penn's Taylor said Team PLUTO decided to make its legged and flying systems completely autonomous. "Once they are launched, they go into the mine and make all the decisions themselves. Like where to go, how to avoid obstacles, and when to communicate. They detect the artefacts on their own, and relay that information back to the base station. Once the robots are launched, the base station operator's only job is to look at what comes back and check it off before it goes back to DARPA. "

Humbert of Team MARBLE agreed: "Once you escape line of sight, you really don't have any way to communicate with the base station. What we've done to compensate for that is to build really strong autonomy into our platform."

Closely related to autonomy, networking and communications were cited by several teams as the biggest challenges of the competition, because radio waves don't go through rock. Again, each team seemed to have a different way of handling this. "We have a hybrid approach," said Hollinger of Team Explorer, "where we allow the vehicles to move out of communication with the operator and come back in. That's where the autonomy kicks in. The operator also has the ability to control and guide the vehicles if they are within communications range." Team Explorer's robots dropped communications nodes in the mine to develop an ad hoc communications network.

GLOSSARY

Ad hoc	eigens zu diesem Zweck, aus dem Stegreif	drop-off	hier: Absatz, Abfall
avoid, to	vermeiden, umgehen	line of sight	Sichtlinie
blimp	Prallluftschiff	node	Netzknoten
boast, to	aufweisen, rühmen	relay, to	weitergeben, übertragen
cite, to	anführen, zitieren	treaded	mit Gleisketten
constraint	Einschränkung, Begrenzung	unique	einzigartig, einmalig

Humbert of Team MARBLE said their robots also established networks by dropping beacons with tiny routers. "These beacons just fall off the vehicles," so the robots don't have to come all the way back to communicate with the base station. "This might save 10-15 minutes over the run," he said.

Team PLUTO, for its part, didn't drop nodes because, according to Taylor, "we are focused on full autonomy." Each of its robots was "outfitted with a mesh networking system that allowed them to connect with each other when they were in communication range. We also built a distributed database system that allows the robots to share information both with each other and the base station."

One of the more unique approaches came from Team CRETISE. Its treaded robot, Cobra, carried miniature treaded vehicles called "First Looks" into the mines. "We drive the Cobra in as far as we can," said Weatherwax, "then it drops the First Looks to maintain a communications link. These nodes are dropped off the Cobra and have the ability to autonomously get themselves right-side up and mobile.

SENSING INSPIRATION

There was broad general agreement about the sensors the teams deployed. Because weight is more of a concern on the drones, however, harder choices had to be made. In every case, the sensors deployed included some combination of "time of flight" cameras to look underneath the robot, infrared to help detect objects, inertial odometry to help gauge the motion of the platform, and LiDAR systems used for range sensing.

"Using all of this camera data," said David Weatherwax of Team CRETISE, "we can detect the different artefact types that have been specified. We know there are only a handful of artefact types we're looking for, so we're training a neural network to detect those artefacts autonomously. By using the different sensors on board the robots, we can figure out where those artefacts exist in space and transmit that information back to DARPA during the run itself." The AI does the initial detection, and the human operator matches what he is seeing.

"While there's wonderful technology here at the challenge," said Humbert of Team MARBLE, "I think the biggest impact is going to be on the students who are coming out of this program. We're training a new generation of autonomous and field roboticists. Sure, we can go out in the lab and drive our little robots down the hallway of the engineering building, and that's good enough for paper. But to have this field experience and to have the students – first, second, and third-year Ph.Ds doing this stuff, is just amazing. In my opinion, this is going to be the largest effect of the challenge."

GLOSSARY

Pair Work.

Partner A: Interviewer Partner B: One of the organizers

B. Drive, Roll, Walk, Fly, Float (3 to 4 Qs)

D. Detection, Sensors (2 to 3 Qs)

READING II

LIVING XENOBOTS LAB REPORT

Read the text and complete the task. There is a glossary at the end of the text.

A book is made of wood. But it is not a tree. The dead cells have been repurposed to serve another need. Now a team of scientists has repurposed living cells - scraped from frog embryos - and assembled them into entirely new life-forms. These millimetre-wide 'xenobots' can move toward a target, perhaps pick up a payload, like a medicine that needs to be carried to a specific place inside a patient, and heal themselves after being cut.

"These are novel living machines," says Joshua Bongard, a computer scientist and robotics expert at the University of Vermont who co-led the new research. "They're neither a traditional robot nor a known species of animal. It's a new class of artefact: a living, programmable organism. "

The new creatures were designed on a supercomputer at UVM and then assembled and tested by biologists at Tufts University. "We can imagine many useful applications of these living robots that other machines can't do," says co-leader Michael Levin, who directs the Center for Regenerative and Developmental Biology at Tufts, "like searching out nasty compounds or radioactive contamination, gathering microplastic in the oceans, traveling in arteries to scrape out plaque."

With months of processing time on a supercomputer cluster, the team used an evolutionary algorithm to create thousands of candidate designs for the new life-forms. Attempting to achieve a task assigned by the scientists, like locomotion in one direction, the computer would, over and over, reassemble a few hundred simulated cells into myriad forms and body shapes. As the programs ran - driven by basic rules about the biophysics of what single frog skin and cardiac cells can do - the more successful simulated organisms were kept and refined, while failed designs were tossed out. After a hundred independent runs of the algorithm, the most promising designs were selected for testing.

Then the team at Tufts transferred the designs into life. First, they gathered stem cells, harvested from the embryos of African frogs, the species *Xenopus laevis* - hence the name 'xenobots'. These were separated into single cells and left to incubate. Then, using tiny forceps and an even tinier electrode, cells were cut and joined under a microscope into a close approximation of the designs specified by the computer.

Assembled into body forms never seen in nature, the cells began to work together. The skin cells formed a more passive architecture, while the once-random contractions of heart muscle cells were put to work creating ordered forward motion as guided by the computer's design, and aided by spontaneous self-organizing patterns allowing the robots to move on their own.

"These xenobots are fully biodegradable," say Bongard, "when they're done with their job after seven days, they're just dead skin cells."

www.uvm.edu/uvmnews/news/team-builds-first-living-robots (2020)

GLOSSARY

forceps	Pinzette	plaque	Plaque, Belag
myriad	Unzahl, Heerschar	repurpose, to	umfunktionieren
nasty	Lästig, übel, böse	scrape, to	schaben, kratzen
novel	neuartig	stem cell	Stammzelle
payload	Traglast	toss out, to	wegwerfen

For questions 1-5, choose the correct answer (A, B, C or D).

1. Which of the following is NOT true about xenobots?

- A They can heal themselves.
- B They can move directionally.
- C They can reproduce themselves.
- D They can transport substances.

2. Xenobots are

- A known creatures.
- B robots.
- C animals.
- D none of the above.

3. The areas of application for xenobots include (more than one answer can be chosen)

- A medicine.
- B dealing with hazardous materials.
- C cleaning up pollution.
- D military use.

4. What does the writer imply about the process of creating xenobots?

- A Xenobots were designed solely by a computer algorithm.
- B The process was similar to the natural evolution.
- C All the simulated organisms were transformed into xenobots.
- D Scientists passively watched the supercomputer cluster at work.

5. Which best describes xenobots' behaviour?

- A they are unable to work cooperatively.
- B they are unable to perform tasks.
- C they are passive.
- D they can move on their own.

RECORDING FEELINGS

Read the text and complete the tasks. There is a glossary at the end of each part of the text.

A

Only your closest friends might be able to detect how you feel from the sound of your voice alone. Are you sure? Big brother is listening now: artificial intelligence is getting better and better at analysing feelings from speech. The coming voice-profiling revolution will completely change how we talk to hotlines - and each other.

You decide to call a store that sells some hiking boots you're thinking of buying. As you dial in, the computer of an artificial intelligence company hired by the store is activated. It retrieves its analysis of the speaking style you used when you phoned other companies the software firm services. The computer has concluded you are "friendly and talkative." Using predictive routing, it connects you to a customer service agent who company research has identified as being especially good at getting friendly and talkative customers to buy more expensive versions of the goods they're considering.

This hypothetical situation may sound as if it's from some distant future. But automated voice-guided marketing activities like this are happening all the time.

If you hear "This call is being recorded for training and quality control," it isn't just the customer service representative they're monitoring. It can be you, too.

When conducting research for my forthcoming book, "The Voice Catchers: How Marketers Listen In to Exploit Your Feelings, Your Privacy, and Your Wallet," I went through over 1,000 trade magazine and news articles on the companies connected to various forms of voice profiling. I examined hundreds of pages of U.S. and EU laws applying to biometric surveillance. I analysed dozens of patents. And because so much about this industry is evolving, I spoke to 43 people who are working to shape it.

It soon became clear to me that we're in the early stages of a voice-profiling revolution that companies see as integral to the future of marketing.

Thanks to the public's embrace of smart speakers, intelligent car displays and voice-responsive phones - along with the rise of voice intelligence in call centres - marketers say they are on the verge of being able to use AI-assisted vocal analysis technology to achieve unprecedented insights into shoppers' identities and inclinations. In doing so, they believe they'll be able to circumvent the errors and fraud associated with traditional targeted advertising.

Not only can people be profiled by their speech patterns, but they can also be assessed by the sound of their voices - which, according to some researchers, is unique and can reveal their feelings, personalities and even their physical characteristics.

FLAWS IN TARGETED ADVERTISING

B

Top marketing executives I interviewed said that they expect their customer interactions to include voice profiling within a decade or so.

Part of what attracts them to this new technology is a belief that the current digital system of creating unique customer profiles - and then targeting them with personalized messages, offers and ads - has major drawbacks.

A simmering worry among internet advertisers, one that burst into the open during the 2010s, is that customer data often isn't up to date, profiles may be based on multiple users of a device, names can be confused and people lie.

Advertisers are also uneasy about ad blocking and click fraud, which happens when a site or app uses bots or low-paid workers to click on ads placed there so that the advertisers have to pay up. These are all barriers to understanding individual shoppers.

Voice analysis, on the other hand, is seen as a solution that makes it nearly impossible for people to hide their feelings or evade their identities.

BUILDING THE MACHINE THAT LISTENS

C Most of the activity in voice profiling is happening in customer support centres, which are largely out of the public eye.

But there are also hundreds of millions of Amazon Echoes, Google Nests and other smart speakers out there. Smartphones also contain such technology.

All are listening and capturing people's individual voices. They respond to your requests. But the assistants are also tied to advanced machine learning and deep neural network programs that analyse what you say and how you say it.

Amazon and Google - the leading purveyors of smart speakers outside China - appear to be doing little voice analysis on those devices beyond recognizing and responding to individual owners. Perhaps they fear that pushing the technology too far will, at this point, lead to bad publicity.

Nevertheless, the user agreements of Amazon and Google - as well as Pandora, Bank of America and other companies that people access routinely via phone apps - give them the right to use their digital assistants to understand you by the way you sound. Amazon's most public application of voice profiling so far is its Halo wristband, which claims to know the emotions you're conveying when you talk to relatives, friends and employers.

GLOSSARY

application	Einsatz, Anwendung	evolve, to	sich entwickeln, sich entfalten
capture, to	erfassen, aufzeichnen	executive	Führungskraft
circumvent, to	umgehen, verhindern	inclination	Neigung, Vorliebe
click fraud	Klickbetrug	on the verge of	kurz vor
convey, to	z. Ausdruck bringen, vermitteln	purveyor	Anbieter/in, Lieferant/in

current	aktuell, gegenwärtig	request	Anliegen, Anfrage
customer service agent	Kundenbetreuer/in	simmer, to	brodeln, langsam kochen
evade, to	ausweichen, umgehen	surveillance	Überwachung

D Fitness trackers like Amazons Halo may soon record not only your physical condition but your mental state as well.

The company assures customers it doesn't use Halo data for its own purposes. But it's clearly a proof of concept – and a nod toward the future.

The patents from these tech companies offer a vision of what's coming.

In one Amazon patent, a device with the Alexa assistant picks up a woman's speech irregularities that imply a cold through using "an analysis of pitch, pulse, voicing, jittering, and/or harmonicity of a user's voice, as determined from processing the voice data." From that conclusion, Alexa asks if the woman wants a recipe for chicken soup. When she says no, it offers to sell her cough drops with one-hour delivery.

Another Amazon patent suggests an app to help a store salesperson decipher a shopper's voice to plumb unconscious reactions to products. The contention is that how people sound allegedly does a better job indicating what people like than their words.

And one of Google's proprietary inventions involves tracking family members in real time using special microphones placed throughout a home. Based on the pitch of voice signatures, Google circuitry infers gender and age information - for example, one adult male and one female child - and tags them as separate individuals.

The company's patent asserts that over time the system's "household policy manager will be able to compare life patterns, such as when and how long family members eat meals, how long the children watch television, and when electronic game devices are working - and then have the system suggest better eating schedules for the kids, or offer to control their TV viewing and game playing.

SEDUCTIVE SURVEILLANCE

E In the West, the road to this advertising future starts with firms encouraging users to give them permission to gather voice data. Firms gain customers' permission by enticing them to buy inexpensive voice technologies.

When tech companies have further developed voice analysis software - and people have become increasingly reliant on voice devices - I expect the companies to begin widespread profiling and marketing based on voice data. Hewing to the letter if not the spirit of whatever privacy laws exist, the companies will forge ahead into their new incarnations, even if most of their users joined before this new business model existed.

This classic bait and switch marked the rise of both Google and Facebook. Only when the numbers of people flocking to these sites became large enough to attract high-paying advertisers did their business models solidify around selling ads personalized to what Google and Facebook knew about their users.

By then, the sites had become such important parts of their users' daily activities that people felt they couldn't leave, despite their concerns about data collection and analysis that they didn't understand and couldn't control.

This strategy is already starting to play out as tens of millions of consumers buy Amazon Echoes at giveaway prices.

THE DARK SIDE OF VOICE PROFILING

F Here's the catch: It's not clear how accurate voice profiling is, especially when it comes to emotions.

It is true, according to Carnegie Mellon voice recognition scholar Rita Singh, that the activity of your vocal nerves is connected to your emotional state. However, Singh told me that she worries that with the easy availability of machine-learning packages, people with limited skills will be tempted to run shoddy analyses of people's voices, leading to conclusions that are as dubious as the methods.

She also argues that inferences that link physiology to emotions and forms of stress may be culturally biased and prone to error. That concern hasn't deterred marketers, who typically use voice profiling to draw conclusions about individuals' emotions, attitudes and personalities.

While some of these advances promise to make life easier, it's not difficult to see how voice technology can be abused and exploited. What if voice profiling tells a prospective employer that you're a bad risk for a job that you covet or desperately need? What if it tells a bank that you're a bad risk for a loan? What if a restaurant decides it won't take your reservation because you sound low class, or too demanding?

Consider, too, the discrimination that can take place if voice profilers follow some scientists' claims that it is possible to use an individual's vocalizations to tell the person's height, weight, race, gender and health.

People are already subjected to different offers and opportunities based on the personal information companies have collected. Voice profiling adds an especially insidious means of labelling. Today, some states such as Illinois and Texas require companies to ask for permission before conducting analysis of vocal, facial or other biometric features.

But other states expect people to be aware of the information that's collected about them from the privacy policies or terms of service - which means they rarely will. And the federal government hasn't enacted a sweeping marketing surveillance law.

With the looming widespread adoption of voice analysis technology, it's important for government leaders to adopt policies and regulations that protect the personal information revealed by the sound of a person's voice.

One proposal: while the use of voice authentication - or using a person's voice to prove their identity - could be allowed under certain carefully regulated circumstances, all voice profiling should be prohibited in marketers' interactions with individuals. This prohibition should also apply to political campaigns and to government activities without a warrant.

That seems like the best way to ensure that the coming era of voice profiling is constrained before it becomes too integrated into daily life and too pervasive to control.

Joseph Turow

This article originally appeared in 'The Conversation' on 28.04.21. It is reproduced here under its CC license.

<https://theconversation.com/shhhh-theyre-listening-inside-the-coming-voice-profiling-revolution-158921>.

Retrieved on 28.02.2022.

GLOSSARY

abuse, to	missbrauchen, falsch anwenden	entice, to	verführen, locken
allegedly	angeblich, vermeintlich	flock, to	strömen, in Scharen kommen
bait and switch	Lockvogeltaktik	hew, to	sich an etwas halten
circuitry	Schaltkreis, Schaltung	infer, to	ableiten, folgern
contention	Behauptung, Streit	proprietary	(urheberrechtlich) geschützt
cough drop	Hustenbonbon	shoddy	schlampig, nachlässig
covet, to	begehren	sweeping	weitreichend, umfassend
decipher, to	entziffern, entschlüsseln	terms of service	Nutzungsbedingungen

Find words in the text with the same or similar meanings as the following.

1. to calculate the amount or value of something (part A) -
2. to open a computer file or to use a computer system such as the internet (part C) -
3. to suddenly make a lot of progress with something (part E) –
4. a sales technique in which goods are advertised at low prices in order to attract customers, although only a small number of the low-priced goods are available (part E) -
5. likely or liable to suffer from (part F) –
6. prevent the occurrence of; discourage (someone) from doing something by instilling doubt or fear of the consequences (part F) –
7. (of a person) making others work hard or meet high standards; not easily satisfied (part F) -
8. proceeding in a gradual, subtle way, but with very harmful effects (part F) –
9. severely restrict the scope, extent, or activity of (part F) –
10. (especially of an unwelcome influence or physical effect) spreading widely throughout an area or a group of people (part F) –

GRAMMAR:

In part E of the text above (SEDUCTIVE SURVEILLANCE) you read the following sentence:

Only when *the numbers of people flocking to these sites became large enough to attract high-paying advertisers* **did their business models solidify** around selling ads personalized to what Google and Facebook knew about their users.

Inversion is used here in order to make the sentences more emphatic. Inversion is more common in academic and journalistic writing than in spoken English.

The normal verb order in a sentence is **noun + verb** but in some emphatic sentences the word order is similar to that in questions: **auxiliary/modal verb + subject + main verb**.

Inversion often happens after the following phrases: only when, only after, never (before), not until, not only...but, no sooner...than, at no time, in no way.

Not only did Isaac Newton **lay** the foundations for **classical mechanics**, **but** he also **discovered** the law of gravity.

TASK: Come up with similar sentences containing inversion. The following prompts might help you. You can also come up with your own ideas.

1. Albert Einstein – developed general theory of relativity – mass-energy equivalence formula $E = mc^2$.

2. Galileo Galilei – telescopic confirmation of the phases of Venus – the discovery of the four largest satellites of Jupiter – established that the sun was at the center of the solar system.

3. Leonardo da Vinci – produced designs for the parachute, helicopter and tank – mapped the human body – his art (the Mona Lisa, The Last Supper).

4. Archimedes – developed theorems to work out the area of a circle, the surface area and volume of a sphere – derived an accurate approximation of pi.

5. Nicolaus Copernicus – formulated a model of the universe that placed the sun rather than the Earth at the center.

WORKPLACE SAFETY

Hazard Pictograms¹

Exercise 1.

Look at the pictograms. Now work in pairs and answer the following questions.

1. Where do you find them?

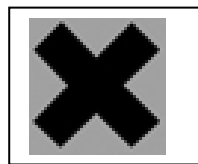
.....

2. What color are they?

.....

3. What is their function?

.....



Exercise 2.

Match the English words with their German equivalent. The first one has been done for you.

English

1 toxic

2 corrosive

3 flammable

4 explosive

5 irritant

6 oxidizing

7 nature polluting

8 refrigerated gas

German

entzündlich

reizend

umweltgefährlich

1 giftig

explosiv

ätzend

brandfördernd

tiefkaltes Gas

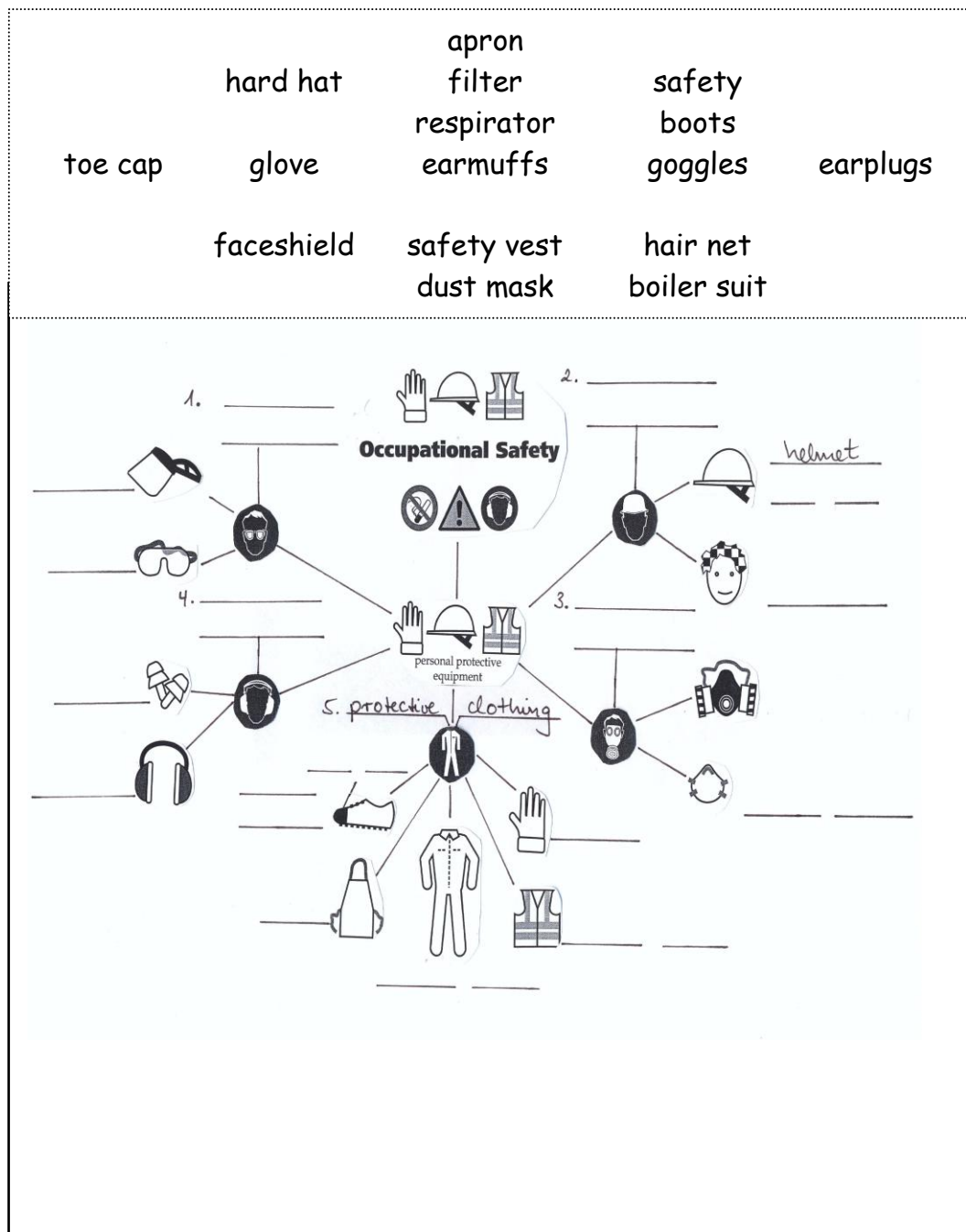
¹ Source reference pictograms and classifying information: <http://echa.europa.eu/web/guest/chemicals-in-our-life/clp-pictograms>

Exercise 3.

Look at the table below and match the information with the corresponding pictogram. The first one is done for you.

New		Information		Information
1		1 a <u>Hazardous to the Environment</u> Very toxic to aquatic life with long lasting effects Toxic to aquatic life with long lasting effects	6	f <u>Corrosive</u> May be corrosive to metals Causes severe skin burns and eye damage
2		b <u>Flammable</u> Extremely flammable gas Flammable gas Extremely flammable aerosol Flammable aerosol Highly flammable liquid and vapour Flammable liquid and vapour Flammable solid	7	g <u>Gas under pressure</u> Contains gas under pressure. May explode if heated. Contains refrigerated gas. May cause burns and injuries.
3		c <u>Serious Health Hazard (chronic toxicity)</u> May be fatal if swallowed Causes damage to organs May damage fertility or the unborn child May cause cancer	8	h <u>Oxidising</u> May cause or intensify fire; oxidiser. May cause fire or explosion; strong oxidiser
4		d <u>Explosive</u> Unstable explosive Explosive; mass explosion hazard Explosive; severe projection hazard Explosive; fire, blast	9	i <u>Health Hazard</u> May cause respiratory irritation May cause drowsiness or dizziness May cause an allergic skin reaction Causes serious eye irritation Harmful if inhaled; Harms public health and the environment by destroying ozone in the upper atmosphere
5		e <u>Acute toxicity</u> Fatal if swallowed Fatal in contact with skin Fatal if inhaled Toxic: if swallowed		

Exercise 4. Complete the pictorial below with the words in the box. You need 'protection' **four** times.



Exercise 5.

Access www.inchbyinch.de/category/pictorial

Type in 'workplace safety', study pictorials on the topic, find 3 words and their German translation. Write them on 3 cards: one side – the English word; the other side – the German translation. Tell your neighbor what they are and learn what words s/he has come up with.

² Source reference: Pictorial – Workplace Safety adapted from: engine – Englisch für Ingenieure (2/2020, pp. 20-21)

.....

.....

.....

Exercise 6³.

Look at pictures A, B and C. Use items from the pictorial in exercise 4 to describe how safety equipment can help to protect the worker.



Example: Carpenter

Read these examples:

1. Goggles **protect** the eyes **from** injuries.
2. Goggles **prevent** the eyes **from being injured**.
3. Goggles **prevent** eye injuries.



Pict. A: Laboratory Assistant



Pict. B: Welder working at steel chain.



Pict.C: Construction Worker

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

³ Source reference: <http://office.microsoft.com/en-us/images/results.aspx?qu=carpenter&ex=2#ai:MC900388960>, <http://office.microsoft.com/en-us/images/results.aspx?qu=welder&ex=1#ai:MC900388968>, <http://office.microsoft.com/en-s/images/results.aspx?qu=protective+clothing&ex=1#ai:MC900359029>, <http://office.microsoft.com/en-us/images/results.aspx?qu=worker&ex=1#ai:MC900297781>